



Subjective Part
(To be solved on Answer Books only)

Subject: Introduction to Software Engineering
Class: BSCYS-II
Section(s): A & B
Course Code: SE101

Time Allowed: 120 Minutes
Max Marks: 50
FM's Name: Muhammad Jalal Shah
FM's Signature:

INSTRUCTIONS

- Attempt responses on the answer book only.
- Nothing is to be written on the question paper.
- Rough work or writing on question paper will be considered as use of unfair means.
- Tables / calculators are not allowed.

Q. No. 1 (CLO 1)		25 Marks
A	Scenario: "A startup is proposing to develop a mobile application that uses AI to provide personalized dietary recommendations based on user-submitted photos of their meals. They are considering three types of feasibility studies: technical, economic, and operational. • Define each of these three types of feasibility studies and explain their importance in the context of this project.	10
	Discuss the key quality attributes of a well-written SRS document. Explain how each attribute contributes to the success of the software development project	5
	Explain the steps to determine the scope of the project and it affects the overall success of the project?	5
Q. No. 2 (CLO 4)		15 Marks
A	Draw a System Use Case (or Use Case Model) for "Halal Verifier," an android application that differentiates between halal and haram products through a barcode scan. To scan a product user has to register and log in to the application. After login, the application will first ask the user to add their allergies if he or she is allergic to some ingredients; after that user will scan the product, and if the product is haram, the application will show him alternative halal products in the same category. In case the product is not harm the application will check for the allergies and will show non allergic products to the user.	15

b	Draw an activity diagram for the scenario mentioned in Part a	5
	Q. No. 3 (CLO 1)	10 Marks

a	A software house is developing a secure medical device for “Remote Patient Monitoring”. Read the scenario below and explain which process model is a good fit for making that device. Scenario: A medical device company is developing a new implantable cardiac monitor for patients with heart conditions. This device will continuously monitor vital signs and transmit data wirelessly to a secure cloud platform accessible by both patients and healthcare professionals. The success of this product hinges on its: • High precision: The device's readings must be accurate and reliable to ensure patient safety and avoid misdiagnosis. • Security: The device and data transmission must be secure to protect sensitive patient health information from unauthorized access. • Usability: The device needs to be user-friendly for patients, potentially with limited technical knowledge, to wear and interact comfortably. Challenges: • Balancing Security and Usability: Implementing robust security measures can add complexity, potentially hindering user experience. • Meeting Regulatory Requirements: The device needs to comply with stringent medical device regulations for safety and efficacy. • Unforeseen Technical Challenges: Unforeseen technical issues related to wireless communication, battery life, or biocompatibility might arise during development.	10
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